

Maths Class: _____

2004

Time allowed: 3 hours plus 5 mins reading time

- Write your name and class at the top of this page, and at the top of each answer sheet.
- At the end of the examination this examination paper must be attached to the front of your answers.
- All questions are of equal value and may be attempted.
- All necessary working must be shown. Marks will be deducted for careless or badly arranged work.
- Marks indicated are a guide only and may be varied if necessary.
- Non programmable calculators may be used.

[illegible]

Question 1**Marks**

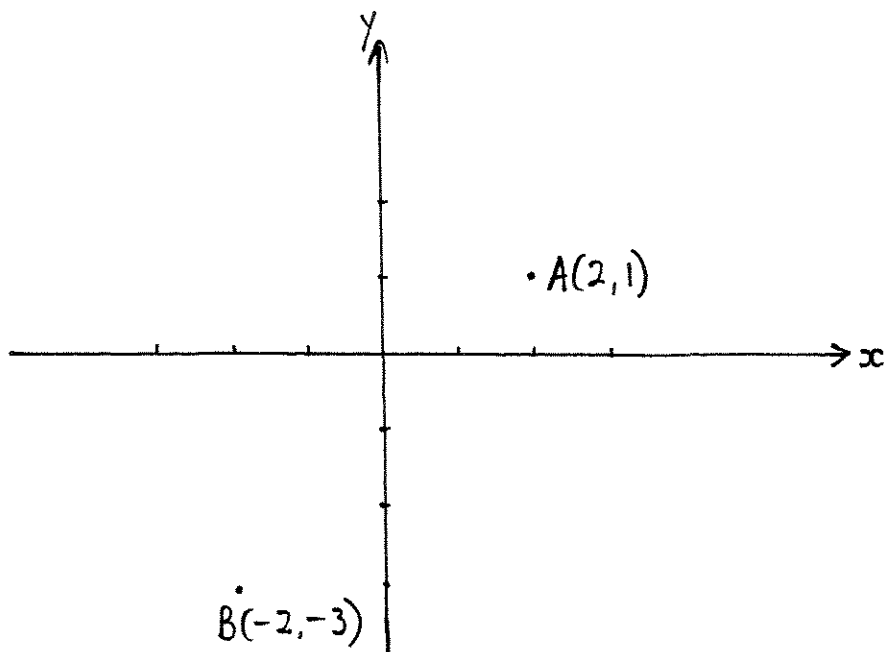
- a) Factorise fully $16x^2 - 81$ 1
- b) Convert $\frac{4\pi}{5}$ radians to degrees 1
- c) Given $f(x) = 1 - x^3$, find x when $f(x) = 65$ 1
- d) Find the values of a and b if $\frac{1}{2\sqrt{3}-1} = a + b\sqrt{3}$ 2
- e) Find the exact value of $\tan 300^\circ$ 2
- f) Evaluate $\lim_{x \rightarrow -2} \frac{3x^2 + 7x + 2}{x + 2}$ 2
- g) Solve and graph the solution on a number line: $|6x - 9| > 21$ 3

Question 2 (Begin a new page)

- a) The roots of the quadratic equation $3x^2 + 4x + 2 = 0$ are α and β .
Find the value of $2\alpha\beta^2 + 2\alpha^2\beta$ 3

b)

Marks



- | | | |
|-------|---|---|
| (i) | Show that the distance between A and B is $4\sqrt{2}$ units. | 2 |
| (ii) | Find the mid-point C, of AB | 1 |
| (iii) | Show that the gradient of AB is 1 | 1 |
| (iv) | Show that the line through C perpendicular to AB has equation $x + y + 1 = 0$ | 2 |
| (v) | Show that this line passes through D (-3, 2) | 1 |
| (vi) | Find the area of $\triangle ABD$ | 2 |

Question 3 (Begin a new page)

a) Differentiate the following with respect to x :

- | | | |
|-------|------------------|---|
| (i) | $x^2 + \sqrt{x}$ | 1 |
| (ii) | $x^2 \tan x$ | 2 |
| (iii) | $\sin(e^x)$ | 2 |

- | | | | |
|----|----------|-------------------------------|---|
| b) | Find (i) | $\int \frac{x^2}{x^3 - 2} dx$ | 2 |
| | (ii) | $\int e^{3x} dx$ | 1 |

c) Evaluate

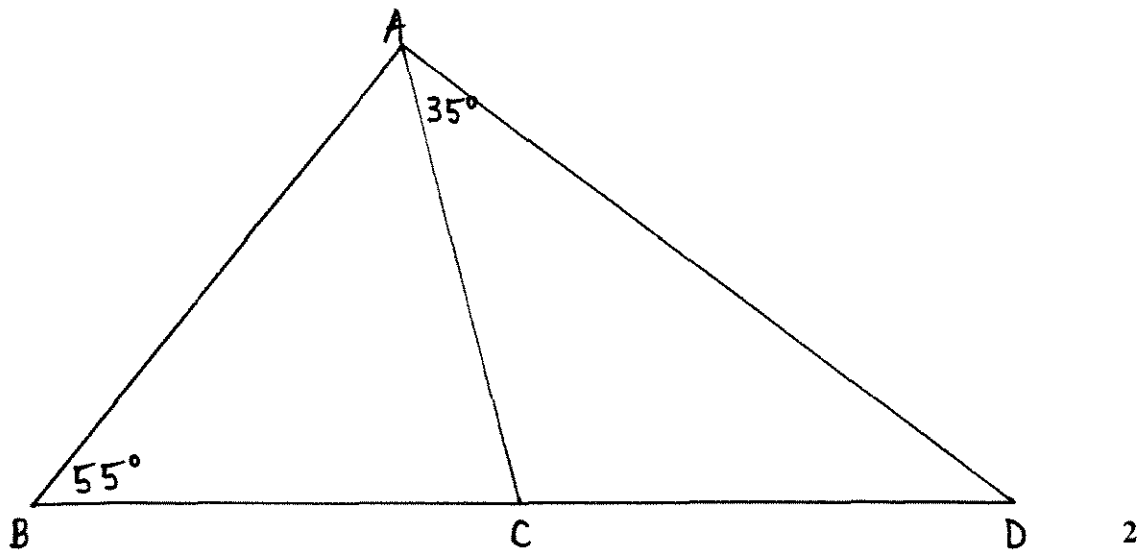
(i) $\int_0^1 (2x+1)^5 dx$ 2

(ii) $\int_0^{\frac{\pi}{4}} \sin 2x dx$ 2

Question 4 (Begin a new page)

a) Find the values of k for which $x^2 + kx + 16$ is positive definite 2

b)



Given that $AC = DC$, $\angle ABC = 55^\circ$ and that $\angle DAC = 35^\circ$, show that triangle ABC is isosceles.

c) Consider the curve whose equation is $y = x^3 - 12x + 5$.

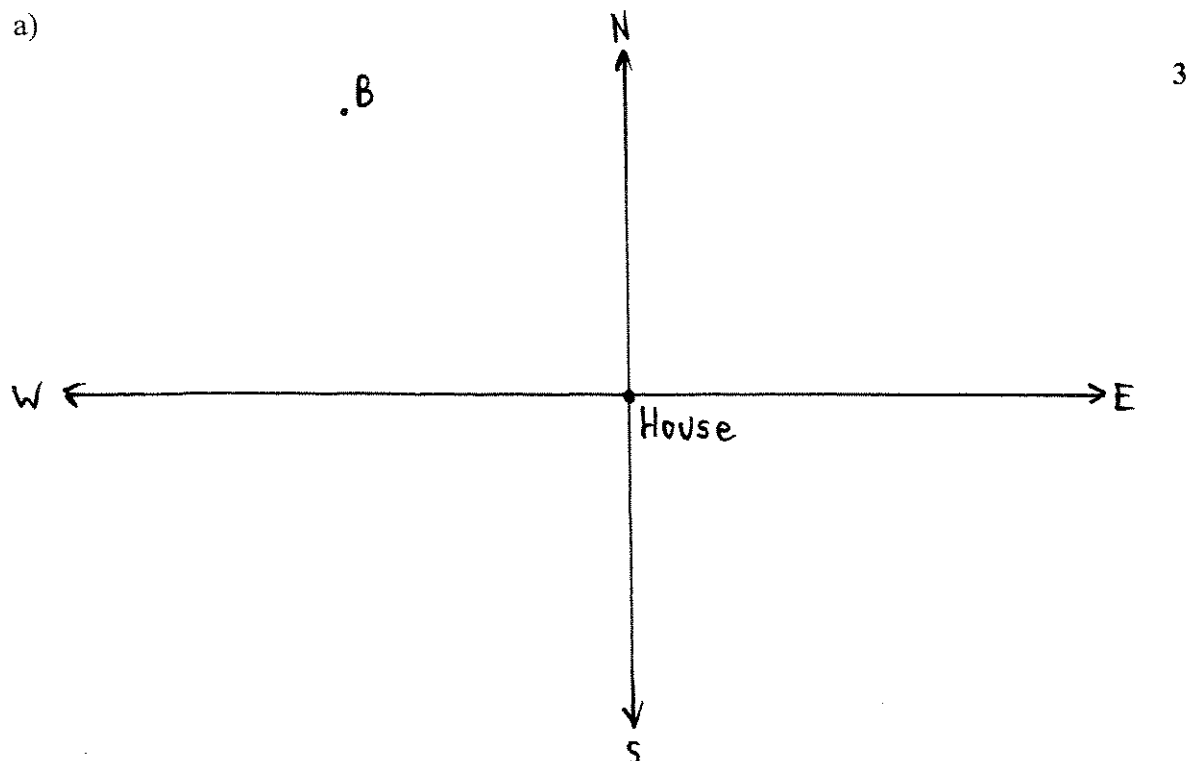
(i) Find the coordinates of the stationary points. 3

(ii) Determine the nature of the stationary points. 1

(iii) Find the point of inflexion. 2

(iv) Sketch the curve over the domain $-3 \leq x \leq 3$. (x intercept not required) 1

(v) Find the minimum value of the function over this domain. 1



Samantha walks from her house for 6km, on a bearing of 310° to point B. She then walks on a bearing of 215° until she is due west of the house. How far is she now from her house? (correct to one decimal place).

- b) The number, N , of people with flu is increasing over time t . Also, the rate at which people are catching flu is increasing.

(i) State the sign (+ or -) of $\frac{dN}{dt}$ and $\frac{d^2N}{dt^2}$ 1

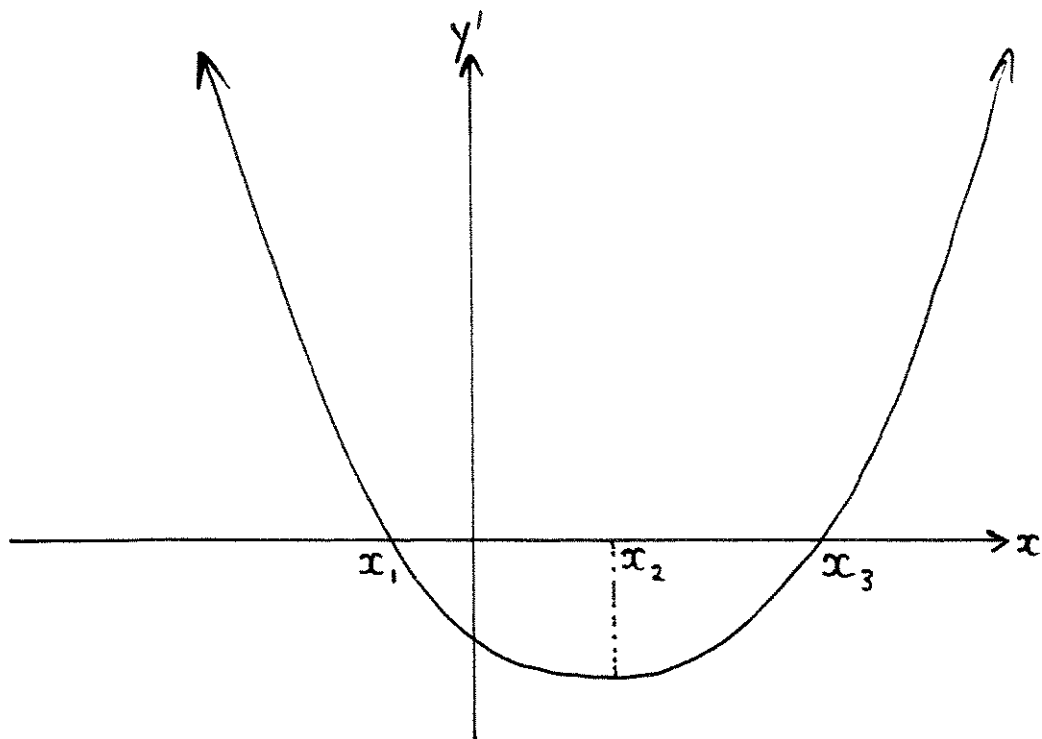
(ii) Sketch a possible graph of $N = f(t)$ which illustrates this information. 1

- c) Given that $\tan A = P$, and $180^\circ < A < 270^\circ$, find an expression for $\cos A$ in terms of P . 2

- d) If $\int_0^a (4 - 2x)dx = 4$, find the value of a . 2

- e) Find the x value of the point on the parabola $y = x^2 + x - 1$ where the tangent is parallel to the line $y = 9x - 5$. 2

f)



1

The sketch above shows the derivative function for a certain curve. Copy this diagram into your answer booklet and on it, sketch a curve that could be the original function.

Question 6 (Begin a new page)

- a) Consider the parabola with equation $x^2 = 8(y - 2)$.
- (i) Find the coordinates of the vertex 1
 - (ii) Find the coordinates of the focus 1
 - (iii) Find the exact volume of the solid formed (a paraboloid) if the portion of the parabola from $y = 2$ to $y = 4$ is rotated about the y axis. 2
- b) To what sum will \$ 4500 amount if invested at 10% p.a. for 6 years if the interest is compounded quarterly? 2

c)

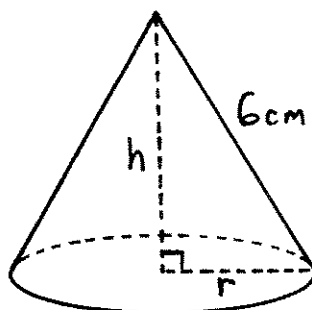
t	0	5	10	15	20
T	83	74	63	50	41

2

The table above shows the temperature T° of an object cooling down over t minutes.

If $T = f(t)$, use all the values in this table, to approximate $\int_0^{20} f(t) dt$ with the Trapezoidal Rule.

d) The slant edge of a right circular cone of height ' h ' and base radius ' r ' cm, is 6cm.



(i) Write down an equation linking r and h . 1

(ii) Given that the formula for the volume of a cone is $V = \frac{1}{3}\pi r^2 h$, 1

use part (i) or otherwise to show $V = 12\pi h - \frac{1}{3}\pi h^3$.

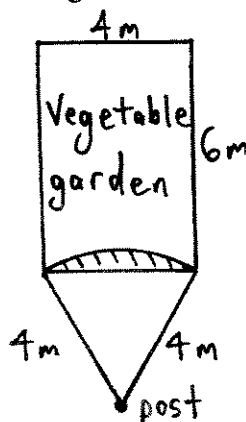
(iii) Hence find the height of the cone which gives a maximum volume. 2

Question 7 (Begin a new page)

a) Solve $25^k (5^3)^4 = 1$ 2

b) A goat is tethered to a 4 metre long rope. The other end of the rope is tied to a post fixed at a point 4 metres from each of two corners of a 6 metre by 4 metre rectangular vegetable garden. This information is illustrated in the diagram below. 3

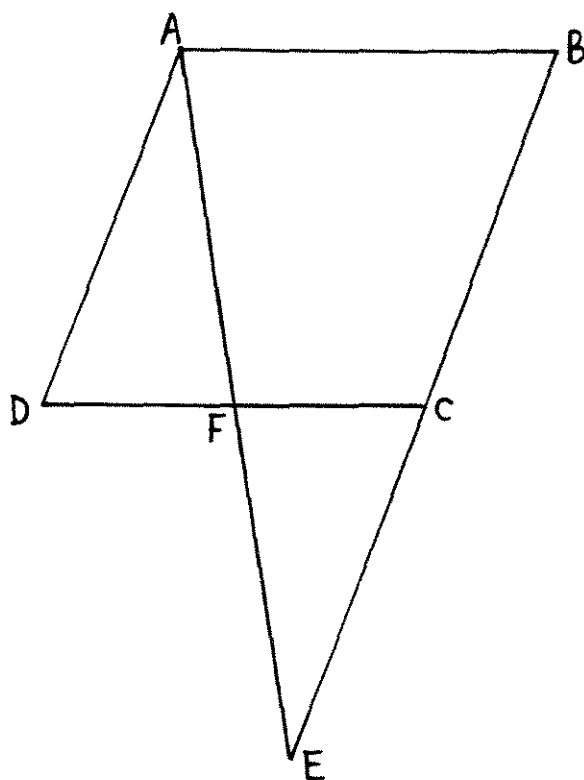
Calculate the exact area of vegetables that the goat can eat.



- c) Evaluate $\sum_{n=3}^{n=12} (2 \times 3^n)$ leaving your answer in index form. 2
- d) (i) Draw a neat sketch of the graph of: $f(x) = -2 \sin x$ for $-\pi \leq x \leq \pi$ 2
- (ii) Show that it is an odd function. 1
- (iii) Hence or otherwise calculate the area bounded by the above curve, the x -axis and between $x = -\pi$ and $x = \pi$. 2

Question 8 (Begin a new page)

a)



The figure above shows a rhombus ABCD with BC produced to E so that $BC = CE$.

Copy this diagram onto your answer page

- (i) Prove that triangles ADF and EBA are similar. 2
- (ii) Prove that F is the midpoint of DC. 2

- b) A chemical substance being made in a laboratory decomposes and the amount M in kilograms present at any time t hours is given by $M = M_0 e^{-kt}$.

If $\frac{3}{4}$ of the mass of this substance will disintegrate in 4 hours, find:

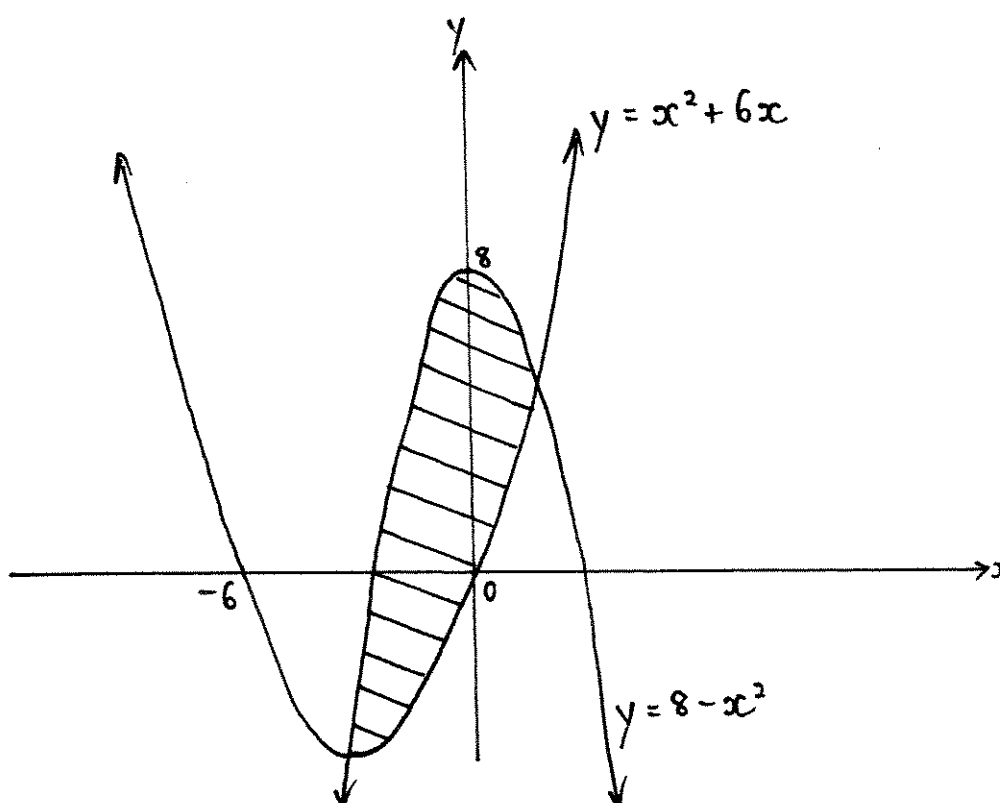
- (i) the value of k , correct to two decimal places. 2
- (ii) the value of M_0 given 4kg of the substance remains after 90 minutes, correct to two decimal places. 2

- c) A particle moving in a straight line with a constant acceleration of 6 m/s^2 , is initially at $x = 2$ with a velocity of 2 m/s .

- (i) Calculate its velocity and displacement in terms of t . 2
- (ii) Draw a velocity time graph for the first four seconds. 1
- (iii) Hence or otherwise find the total distance travelled during the first 4 seconds. 1

Question 9 (Being a new page)

a)



Calculate the area of the shaded region above.

- b) The second term of a geometric series is 27 and the fifth term is 64.
- (i) Find the first term and the common ratio. 2
 - (ii) Find the sum of the first five terms of this series. 2
- c) Show that if $y = \ln \left(\frac{2x+1}{3x-1} \right)$, then $\frac{dy}{dx} = -\frac{5}{(2x+1)(3x-1)}$ 2
- d) Solve $\cos^2 2x = \frac{1}{4}$ for $0 \leq x \leq 360^\circ$ 3

Question 10 (Begin a new page)

- a) A square metal plate, with an original side length of 20cm, is being heated so that the length 'L' of each side of the plate at any time 't' is
- $$L = 4t + 20.$$
- (i) Find an expression for the area of the plate at time 't' seconds. 1
 - (ii) After what time has the area of the plate reached 784cm^2 ? 2
 - (iii) Find the rate of increase of the area when $t = 1$ second. 2
- b) Bill borrows \$100000 at 6% p.a. monthly reducible, to be repaid monthly over 10 years.
- (i) Given he pays \$P per month, and the amount owing after n months is A_n , show that after 2 months, the amount owing is

$$A_2 = 100000(1.005)^2 - P(1 + 1.005)$$
 2
 - (ii) Hence show that the amount owing after n months is:

$$A_n = 100000(1.005)^n - 200P(1.005^n - 1)$$
 3
 - (iii) Calculate to the nearest cent, the monthly repayment required to repay the loan in 10 years. 2

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Question

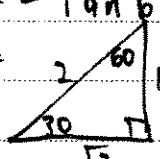
1a) $16x^2 - 81$
 $(4x-9)(4x+9)$ ①

b) $\frac{4\pi}{5} \times \frac{180}{\pi}$
 $= 144^\circ$ ①

c) $f(x) = 1 - x^3$
 $f(x) = 65 = 1 - x^3$
 $x^3 = -64$
 $x = -4$ ①

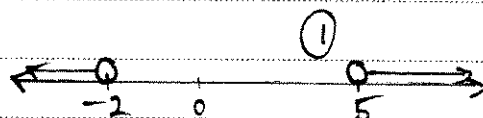
d) $\frac{1}{2\sqrt{3}-1} \times \frac{2\sqrt{3}+1}{2\sqrt{3}+1}$
 $= \frac{2\sqrt{3}+1}{12-1}$
 $= \frac{2\sqrt{3}+1}{11}$
 $= \frac{1}{11} + \frac{2}{11}\sqrt{3}$
 $a = \frac{1}{11}, b = \frac{2}{11}$ ①

e) $\tan 300$
 ① $= -\tan 60$
 $= -\sqrt{3}$ ①



f) $\lim_{x \rightarrow -2} \frac{3x^2 + 7x + 2}{x+2}$
 $= \lim_{x \rightarrow -2} \frac{(3x+1)(x+2)}{x+2}$ ①
 $= -5$ ①

g) $|6x-9| > 21$
 $6x-9 > 21, 6x-9 < -21$
 $6x > 30, 6x < -12$
 $x > 5$ ① $x < -2$ ①



Question

2. a) $3x^2 + 4x + 2 = 0$
 $2 \times \beta^2 + 2 \times \alpha^2 \beta$
 $= 2 \times \beta(\beta + \alpha)$
 $= 2 \times \frac{c}{a} \times \frac{-b}{a}$ ①
 $= 2 \times \frac{2}{3} \times \frac{-4}{3}$ ①
 $= \frac{-16}{9}$ ①

b) i) $AB = \sqrt{(2-(-2))^2 + (1-(-3))^2}$ ①
 $= \sqrt{4^2 + 4^2}$
 $= \sqrt{32}$
 $= 4\sqrt{2}$ units ①

ii) $C = \left(\frac{-2+2}{2}, \frac{-3+1}{2} \right)$
 $= (0, -1)$ ①

iii) $M_{AB} = \frac{1-(-3)}{2-(-2)}$
 $= \frac{4}{4}$
 $= 1$ ①

iv) $\therefore \perp M_{AB} = -1$

$C(0, -1)$

$y - y_1 = m(x - x_1)$ ①

$y - (-1) = -1(x - 0)$

$x + y + 1 = 0$ as required ①

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cv) $D(-3, 2)$ lies on $x+y+1=0$
if it satisfies equation.
 $-3+2+1=0$ ✓ ∴ Yes

①

$$cv) A = \frac{1}{2} b h$$

$$= \frac{1}{2} \times AB \times CD$$

$$= \frac{1}{2} \times 4\sqrt{2} \times CD$$

$$CD = d. bet. (0, -1) \text{ and } (-3, 2)$$

$$= \sqrt{(-3-0)^2 + (2-(-1))^2}$$

$$= \sqrt{9+9}$$

$$= 3\sqrt{2}$$

①

$$A = \frac{1}{2} \times 4\sqrt{2} \times 3\sqrt{2}$$

$$= 12 \text{ units}^2$$

①

Question

$$3. \text{ a) } x^2 + \sqrt{x}$$

$$\frac{d}{dx}(x^2 + x^{\frac{1}{2}})$$

$$= 2x + \frac{1}{2}x^{-\frac{1}{2}} \quad \text{①}$$

$$\text{b) } x^2 \tan x$$

$$\frac{d}{dx} = x^2 \sec^2 x + \tan x \times 2x \quad \text{①}$$

$$= x^2 \sec^2 x + 2x \tan x \quad \text{①}$$

$$\text{c) } \sin(e^x)$$

$$\frac{d}{dx} = \cos e^x \times e^x$$

$$= e^x \cos(e^x) \quad \text{①}$$

$$\text{① a) } \int \frac{x^2}{x^3-2} dx$$

$$= \frac{1}{3} \int \frac{3x^2}{x^3-2} dx \quad \text{①}$$

$$= \frac{1}{3} \ln(x^3-2) + c \quad \text{①}$$

$$\text{b) } \int e^{3x} dx$$

$$= \frac{1}{3} \int 3e^{3x} dx$$

$$= \frac{1}{3} e^{3x} + c \quad \text{①}$$

$$\text{① c) } \int_0^1 (2x+1)^5 dx$$

$$\left[\frac{(2x+1)^6}{6 \times 2} \right]_0^1 \quad \text{①}$$

$$= \frac{3^6 - 1^6}{12}$$

$$= 60 \frac{2}{3} \quad \text{①}$$

$$\text{d) } \int_0^{\frac{\pi}{4}} \sin 2x dx$$

$$\left[-\frac{1}{2} \cos 2x \right]_0^{\frac{\pi}{4}} \quad \text{①}$$

$$= -\frac{1}{2} (\cos \frac{\pi}{2} - \cos 0)$$

$$= -\frac{1}{2} (0 - 1)$$

$$= \frac{1}{2} \quad \text{①}$$

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Question

4 a) $x^2 + kx + 16$

$\Delta < 0$

$k^2 - 4 \times 1 \times 16 < 0$ ①

$k^2 - 64 < 0$

$(k-8)(k+8) < 0$

$-8 < k < 8$ ①

b) Since $\triangle ACD$ is isosceles,

$\angle ACB = 2 \times 35$

 $= 70^\circ$ (exterior angle of a triangle)

$\therefore \angle BAC = 180 - 55 - 70$ ①

 $= 55^\circ$ (Angle sum of a triangle) $\Rightarrow \triangle ABC$ is isosceles as it contains 2 equal angles. ①

c) i) $y = x^3 - 12x + 5$

Stat. pts. where $y' = 0$

$y' = 3x^2 - 12 = 0$ ①

$x^2 - 4 = 0$

$x = \pm 2$

 $\therefore (2, -11)$ ① $(-2, 21)$ ① are the stationary points

ii) $y'' = 6x$

When $x = 2$, $y'' > 0$ $\therefore (2, -11)$ is a minimum

turning pt. ①

When $x = -2$, $y'' < 0$ $\therefore (-2, 21)$ is a maximum

turning pt.

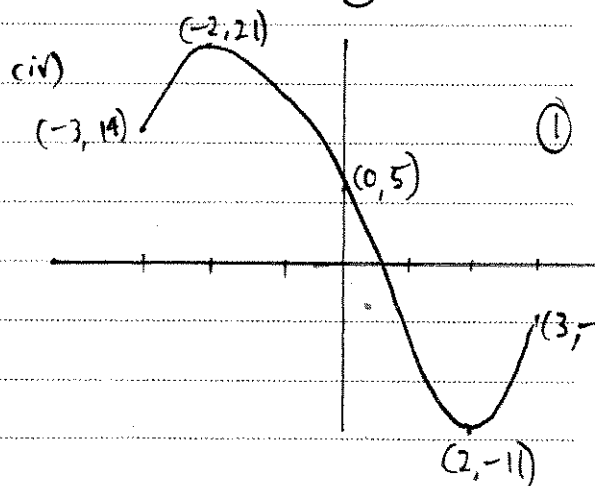
iii) Pt. of inflexion occurs

when $y'' = 0$

$6x = 0$ ①

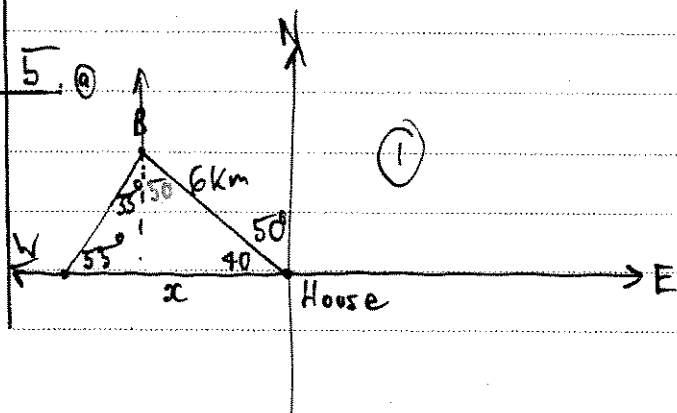
when $x = 0$ must be

a pt. of inflexion (non horizontal)

as $y' \neq 0$ at this point $\therefore (0, 5)$ is the inflexion ①cr) -11 is the minimum value ①

Question

5. a)



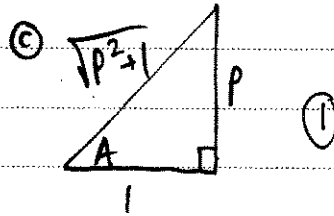
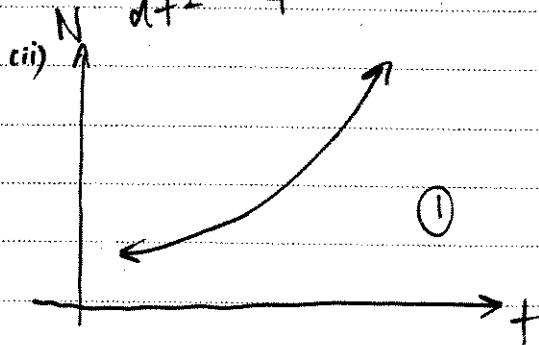
$\frac{x}{\sin 85} = \frac{6}{\sin 55}$ ①

$x = 7.3 \text{ km}$ ①

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(b) (i) $\frac{dN}{dt} +$, $\frac{d^2N}{dt^2} +$ ①



$\cos A = \frac{1}{\sqrt{P^2+1}}$ ① as A is in 3rd quadra.

(d) $\int_0^9 4 - 2x \, dx = 4$

$[4x - x^2]_0^9 = 4$

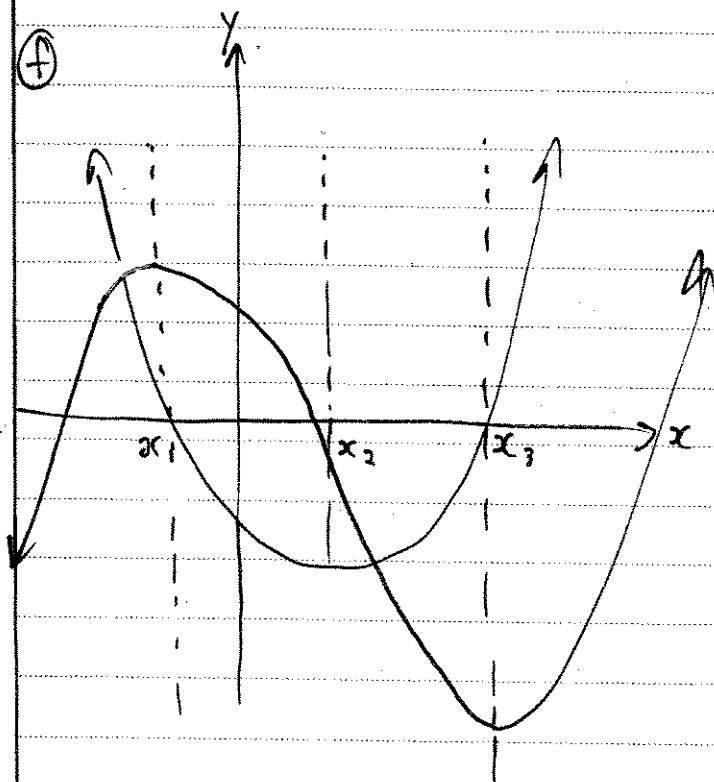
$4a - a^2 - 0 = 4$ ①

$a^2 - 4a + 4 = 0$

$(a-2)^2 = 0$

$a = 2$ ①

(e) $y = x^2 + x + 1$
 $\frac{dy}{dx} = 2x + 1$ parallel to $y = 9x$
 $\Rightarrow 2x + 1 = 9$ ①
 $2x = 8$
 $x = 4$ ①



Stationary pts.
must line up with
 x_1 and x_3 .

①

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Question

6. a) $x^2 = 8(y-2)$
 (i) Vertex $(0, 2)$ ①
 (ii) Focus $(0, 4)$ ①

(iii) $V = \pi \int_2^4 x^2 dy$
 $= \pi \int_2^4 8(y-2) dy$
 $= 8\pi \left[\frac{y^2}{2} - 2y \right]_2^4$ ①
 $= 8\pi [8 - 8 - (2 - 4)]$
 $= 16\pi \text{ units}^3$ ①

④ (i) $r^2 + h^2 = 6^2$
 $r^2 + h^2 = 36$ ①

(ii) $V = \frac{1}{3} \pi r^2 h$
 $= \frac{1}{3} \pi (36 - h^2) \times h$ from (i)
 $V = 12\pi h - \frac{1}{3} \pi h^3$ ①

b) $A = P \left(1 + \frac{r}{100} \right)^n$
 $r = 10 \div 4 = 2.5$
 $n = 6 \times 4 = 24$

$\therefore A = 4500 \left(1 + \frac{2.5}{100} \right)^{24}$ ①
 $= \$8139.27$ ①

⑥ $\int_0^{20} f(t) dt = \frac{5}{2} (83 + 41 + 2(74 + 63 + 50))$ ①
 $= 2\frac{1}{2} \times (124 + 187 \times 2)$
 $= 1245$ ①

(iii) $\frac{dV}{dh} = 12\pi - \pi h^2 = 0$
 for a maximum
 $12 - h^2 = 0$
 $h^2 = 12$
 $h = \sqrt{12}$
 $h = 2\sqrt{3}$ ①
 $\frac{d^2V}{dh^2} = -2\pi h$
 $= -2\pi \times 2\sqrt{3} < 0$
 when $h = 2\sqrt{3} \therefore$ ①
 $h = 2\sqrt{3}$ gives a max. volume

Question

7. a) $25^k (5^3)^4 = 1$
 $(5^2)^k \times 5^{12} = 1$
 $5^{2k+12} = 5^0$ ①
 $\therefore 2k + 12 = 0$
 $k = -6$ ①

b) $A = \frac{1}{2} r^2 (\theta - \sin \theta)$ ① area of seg
 $= \frac{1}{2} \times 4^2 \left(\frac{\pi}{3} - \sin \frac{\pi}{3} \right)$ ① $60^\circ = \frac{\pi}{3}^c$
 $= 8 \left(\frac{\pi}{3} - \frac{\sqrt{3}}{2} \right) \text{ m}^2$ ①

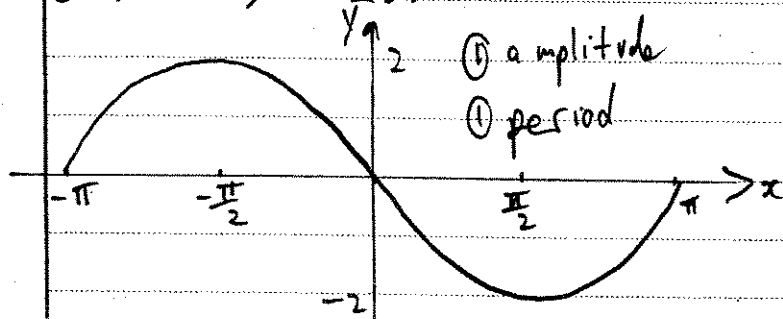
$8 \left(\frac{2\pi - 3\sqrt{3}}{6} \right) = \frac{4(2\pi - 3\sqrt{3})}{3} \text{ m}^2$

⑦ $\sum_{n=3}^{12} (2 \times 3^n) = 54 + 162$
 $= 2(27 + 81 + \dots + \dots)$ ①
 18 terms ①
 $= 2 \times \frac{27(3^{10} - 1)}{2}$ ①
 $= 27(3^{10} - 1)$

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(d) (i) $f(x) = -2\sin x$



① amplitude

① period

(ii) Odd if

$$-f(x) = f(-x)$$

$$-(-2\sin x) = -2\sin(-x)$$

$$2\sin x = -2(-\sin x)$$

$$2\sin x = 2\sin x \quad \checkmark \quad \text{①}$$

 $\therefore$ odd

(iii) $A = 2 \int_{-\pi}^0 -2\sin x \, dx$

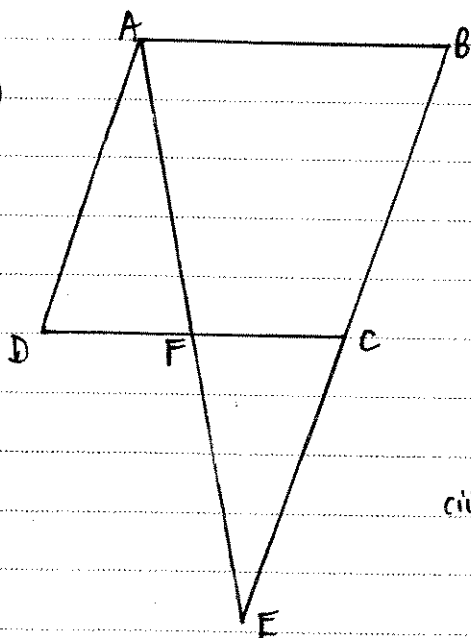
$$= -4 [-\cos x]_{-\pi}^0 \quad \text{①}$$

$$= 4 [\cos 0 - \cos \pi]$$

$$= 4 (1 - (-1))$$

$$= 8 \text{ units}^2 \quad \text{①}$$

Question 8: (a)



(i) In Δ 's ADF and EBA,
 $\angle D = \angle B$ (opposite angles of a rhombus)

$\angle DAF = \angle FEB$ (alternate angles in parallel lines)

$\therefore \Delta ADF \sim \Delta EBA$ (equiangular)

(ii) Since $EB : AD = 2 : 1$ ($EC = BC = 1$)

$\therefore AB : DF = 2 : 1$ (corresponding sides in similar triangle) ①

But $AB = DC$ (opposite sides of rhombus)

$$\therefore DF : DC = 1 : 2 \quad \text{①}$$

$\therefore$ F must be the midpoint of DC.

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⑥ ci) $M = M_0 e^{-kt}$

when $t=0$, $M = M_0$ $\therefore$ initial amt. is M_0

when $t=4$, $M = \frac{M_0}{4}$ ($\frac{3}{4}$ disintegrated)

$$\therefore \frac{M_0}{4} = M_0 e^{-4k}$$

$$\frac{1}{4} = e^{-4k} \quad \text{①}$$

$$\log_e \frac{1}{4} = -4k$$

$$k = \frac{\log_e \frac{1}{4}}{-4}$$

$$k = 0.3465735 \quad \text{①}$$

cii) when $t=90 = 1\frac{1}{2}$ hours, $M = 4\text{kg}$

$$M = M_0 e^{-kt}$$

$$4 = M_0 e^{-0.3465735 \times 1\frac{1}{2}} \quad \text{①}$$

$$M_0 = 6.73\text{kg} \text{ correct to 2 d.p.'s.} \quad \text{①}$$

⑦ ci) $a = 6$

$$v = 6t + C$$

when $t=0$, $v=2$

$$2 = 0 + C$$

$$\therefore C = 2$$

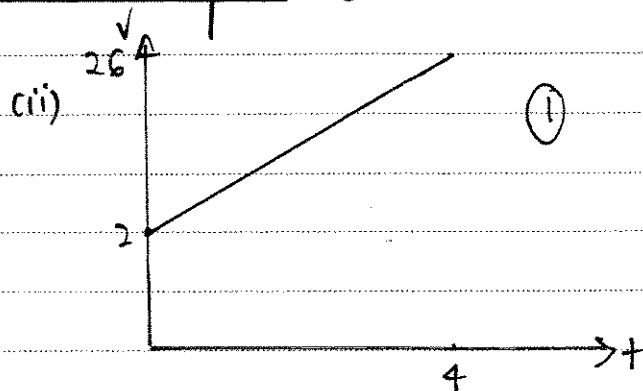
$$\therefore v = 6t + 2 \quad \text{①}$$

$$x = 3t^2 + 2t + C$$

when $t=0$, $x=2$

$$\therefore C = 2$$

$$x = 3t^2 + 2t + 2 \quad \text{①}$$



$$\begin{aligned} \text{ciii) distance} &= \int_0^4 v \, dt \\ &= \text{Area of trapezium} \\ &= \frac{2+26}{2} \times 4 \\ &= 28 \times 2 \\ &= 56\text{m} \quad \text{①} \end{aligned}$$

Question 9 @ $A = \int \text{top curve} - \text{bottom curve} \, dx$

Pts. of intersection at:

$$x^2 + 6x = 8 - x^2$$

$$2x^2 + 6x - 8 = 0$$

$$x^2 + 3x - 4 = 0$$

$$(x-1)(x+4) = 0 \quad \text{①}$$

$$\therefore A = \int_{-4}^1 8 - x^2 - (x^2 + 6x) \, dx$$

$$= \int_{-4}^1 8 - 2x^2 - 6x \, dx$$

$$= \left[8x - \frac{2}{3}x^3 - 3x^2 \right]_{-4}^1 \quad \text{①}$$

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$$\begin{aligned}
 &= 8 - \frac{2}{3} - 3 - \left(8x - 4 - \frac{2}{3}x - 4^3 - 3x - 4^2\right) \\
 &= 4\frac{1}{3} - (-32 + \frac{128}{3} - 48) \\
 &= \underline{41\frac{2}{3} \text{ Units}^2} \quad (1)
 \end{aligned}$$

$$\begin{aligned}
 (b) \quad T_n &= ar^{n-1} \\
 T_2 &= 27 = ar^{2-1} \Rightarrow 27 = ar \quad (1) \\
 T_5 &= 64 = ar^{5-1} \quad 64 = ar^4 \quad (2)
 \end{aligned}$$

$$\begin{aligned}
 \therefore (2) \div (1) \text{ gives } \frac{64}{27} &= r^3 \\
 \therefore r &= \frac{4}{3} \quad (1) \\
 \therefore 27 &= a \times \frac{4}{3} \\
 a &= 20\frac{1}{4} \quad (1)
 \end{aligned}$$

$$\begin{aligned}
 (ii) \quad S_5 &= \frac{a(r^5 - 1)}{r - 1} \\
 &= \frac{20\frac{1}{4} \left(\frac{4}{3}^5 - 1\right)}{\frac{4}{3} - 1} \quad (1) \\
 &= \underline{195.25} \quad (1)
 \end{aligned}$$

$$\begin{aligned}
 (c) \quad y &= \log_e \left(\frac{2x+1}{3x-1} \right) \\
 &= \log_e(2x+1) - \log_e(3x-1) \\
 y' &= \frac{2}{2x+1} - \frac{3}{3x-1} \quad (1)
 \end{aligned}$$

$$\begin{aligned}
 y' &= \frac{2(3x-1) - 3(2x+1)}{(2x+1)(3x-1)} \\
 &= \frac{6x - 2 - 6x - 3}{(2x+1)(3x-1)} \\
 &= \frac{-5}{(2x+1)(3x-1)} \quad (1)
 \end{aligned}$$

$$\begin{aligned}
 (d) \quad \cos^2 2x &= \frac{1}{4} \\
 \cos 2x &= \pm \frac{1}{2} \quad (1) \\
 2x &= 60^\circ, 120^\circ, 240^\circ, 300^\circ, 420^\circ, 480^\circ \\
 &\quad (1) \quad 600^\circ, 660^\circ \\
 x &= 30^\circ, 60^\circ, 120^\circ, 150^\circ, 210^\circ, 240^\circ, 300^\circ, 330^\circ \quad (1)
 \end{aligned}$$

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$$= (4t + 20)^2$$

$$= 16t^2 + 160t + 400 \quad (1)$$

$$(ii) A = 784 = 16t^2 + 160t + 400$$

$$0 = 16t^2 + 160t - 384$$

$$0 = t^2 + 10t - 24 \quad (1)$$

$$0 = (t-2)(t+12)$$

$$\therefore t = 2 \text{ seconds as } t > 0.$$

$$(iii) \frac{dA}{dt} = 32t + 160 \quad (1)$$

$$\therefore \text{when } t = 1,$$

$$\frac{dA}{dt} = 192 \text{ cm}^2/\text{s} \quad (1)$$

$$(b) (i) A_1 = 100000 \times 1.005 - P \quad (1) \text{ as } 6\% \text{ p.a.} = 0.5\% \text{ p/month}$$

$$A_2 = A_1 \times 1.005 - P$$

$$= (100000 \times 1.005 - P) \times 1.005 - P$$

$$= 100000 \times 1.005^2 - P(1 + 1.005) \text{ as req'd} \quad (1)$$

$$(ii) A_n = 100000 \times 1.005^n - P(1 + 1.005 + 1.005^2 + \dots + 1.005^{n-1})$$

C.P. $a=1, r=1.005, n=n$ (1)

$$= 100000 \times 1.005^n - P \frac{(1.005^n - 1)}{1.005 - 1} \quad (1)$$

$$= 100000 \times 1.005^n - \frac{P(1.005^n - 1)}{0.005}$$

$$= 100000 \times 1.005^n - 200P(1.005^n - 1) \text{ as req'd.} \quad (1)$$

$$(iii) \text{After 10 years } n = 120$$

$$\therefore A_{120} = 0 = 100000 \times 1.005^{120} - 200P(1.005^{120} - 1) \quad (1)$$

$$\therefore P = \frac{100000 \times 1.005^{120}}{200(1.005^{120} - 1)}$$

$$P = \$1110.21 \text{ a month} \quad (1)$$

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Student's Name/Nº: